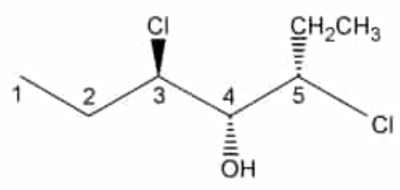
The compound shown below has selected carbons labeled.

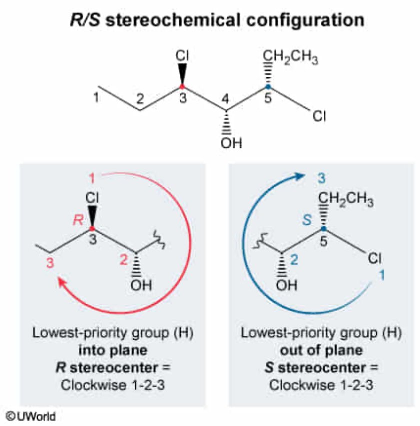


Where are the two stereocenters, with one in the *S* configuration and the other in the *R* configuration, located?

- A. Carbon 3 is *S*, and Carbon 4 is *R*. [4%]
- B. Carbon 3 is *S*, and Carbon 5 is *R*. [7%]
- ✔ C. Carbon 3 is *R*, and Carbon 5 is *S*. [83%]
- D. Carbon 4 is *R*, and Carbon 5 is *S*. [4%]

Correct
83% answered correctly

Explanation:



A **chiral carbon** is a carbon with **four different substituents**. The ***R* or *S*** configuration (stereochemistry) of the stereocenter depends on the spatial arrangement of the substituents. To determine *R/S* configuration:

- The priority of substituents attached to the chiral carbon is assigned, with higher priority given to atoms with greater atomic number. In case of a priority tie, the atomic numbers of the next attached atoms are considered.
- The circular arrangement of groups 1, 2, and 3 is examined. If the **lowest-priority** group (4) is pointing into the plane (**dash**), a **clockwise** circle formed by these groups is ***R*** and a counterclockwise circle is ***S***. If the **lowest-priority** group is pointing out of the plane (**wedge**), a **clockwise** circle is ***S*** and a **counterclockwise** circle is ***R***.

Carbons 3 and 5 are chiral because each has four different substituents. The priority of the substituents on C3 and C5 is, from highest to lowest (1–4): Cl, $CH(OH)CH(CH_2CH_3)Cl$, CH_2CH_3, and H. Priorities 1-2-3 are arranged in a clockwise fashion for both carbons, but the lowest-priority group (H) is pointed into the plane on C3 and out of the plane on C5. Therefore, C3 is in the *R* configuration, whereas C5 is in the *S* configuration.

(Choices A and D) C4 is not chiral because it does not contain four *different* substituents. It is connected to $CH(Cl)CH_2CH_3$ on both sides, making the compound symmetrical.

(Choice B) The *R* and *S* assignments of C3 and C5 are incorrectly reversed.

Educational objective:
The *R* or *S* configuration of a chiral carbon depends on the spatial arrangement of its substituents. When the lowest-priority group is pointed into the plane (dash), the chiral carbon is *R* if priorities 1-2-3 are arranged in a clockwise fashion and *S* if they are counterclockwise. When the lowest-priority group is pointed out of the plane (wedge), the chiral carbon is *S* if priorities 1-2-3 are arranged in a clockwise fashion and *R* if they are counterclockwise.

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